

THE MORPHOLOGICAL, THERMAL AND MECHANICAL PROPERTIES OF GROUND RUBBER TIRE (GRT) FILLED WASTE HIGH-DENSITY POLYETHYLENE (rHDPE)

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ABSTRACT

This research investigates the morphological, thermal and mechanical properties of ground rubber tire (GRT) filled waste high-density polyethylene (rHDPE), showing the effect of variable particle sizes. In the adopted methodology, the samples were prepared through melt mixing (using two-roll mill machine), tailed by compression molding, different blends of variable particle sizes were prepared (150 μm , 212 μm , 300 μm and 425 μm) and a control sample of 150g rHDPE was also prepared. The morphology, thermal properties and mechanical properties of the developed composites are analyzed using Scanning electron microscopy (SEM), Dynamic Mechanical Analysis (DMA), tensile and flexural tests respectively. The SEM result shows some inhomogeneity and poor dispersion of the GTR filler which attributed to improper stress transfer along the interface and formation of cracks in the developed polymer composited, causing reduction in mechanical strength. A more homogenous morphology with reduced defects was observed upon increasing the filler content from 150 μm to 212 μm , leading to an improved interfacial adhesion with the HDPE matrix. Further increase (from 212 μm to 300 μm) gave more homogenous micrograph but addition of GTR particles up 425 μm led to rough surface. For thermal behavior, as the different sizes of the GRT filler were introduced into the HDPE matrix, the storage modulus decreases, attributed to the material softening caused by the soft rubber particles which result in lower rigidity except for sample made with 150 μm . Incorporation of 300 μm particles of GTR results in higher rigidity, which similarly gave more homogenous micrograph with uniform dispersion. The result also shows that, increasing the GTR particles causes decline in the flexural stress and modulus, having similar trend with the tensile modulus. The elongation at break also shows a decline with addition of the different particle sizes of the filler, with the exception of sample 300 μm .

Keywords: Morphological, Ground Rubber Tire (GRT), Waste High-Density Polyethylene (rHDPE), Scanning Electron Microscopy (SEM), Dynamic Mechanical Analysis (DMA)

INTRODUCTION

Composite material are widely used in various industries such as construction, automotive, aerospace, and packaging due to their desirable properties such as high strength, light-weight, and cost-effectiveness. Polymer composite in particular have gained significant attention as they offer

potential of using recycled materials, which are cost effective and eco-friendly. Ground rubber is one of the recycled material which can be used to replace virgin fillers in polymer composites. High-density polyethylene (HDPE) is a popular polymer matrix for composite materials due to its excellent mechanical and thermal properties (Egodage, Harper and Walpalage, 2009).

The thermal and morphological properties of a composite material play a pivotal role in determining its suitability for various application. Thermal properties such as thermal conductivity, thermal diffusivity, and specific heat capacity are important parameter to consider as they affect the heat transfer characteristic of the material also the morphological properties such as surface morphology, microstructure, and particle size distribution are also important as they determine the mechanical and physical properties of the composite material

(Rajalingam, Sharpe and Baker, 2013).

The aim of this research is to analyse the morphological and thermal properties of ground rubber filled with HDPE as ground rubber is one of the recycled materials which can be used to replace virgin fillers in polymer composites if the result come out to be positive.

The objective was to convert both the ground rubber and HDPE into powder of various particulate size (150 μ , 212 μ , 300 μ , 425 μ) and evaluate the morphological properties (Scanning electron microscopy 'SEM') and the thermal properties (Dynamic mechanical analysis 'DMA') alongside the Tensile strength and the Flexural strength.

The research shows that Polymer matrix thermoplastic composites are generally non-biodegradable and are therefore a major contribution to environmental pollution. The difficulty in rubber recycling lies in the fact that rubber is a thermoset material and attempts to recycle it in a manner similar to plastics often lead to non-desired results. The volumes of discarded tires. Ground tire rubbers in low particle sizes would be a good option for this type of application. Although the lack of adhesion of ground tire rubber with the rest of the polymers makes this practice hard also with some problems associated with the storage and elimination of end-of-life tires lead to growing world-wide environmental concerns and problems. This is why there is a need to seeking friendly and eco-friendly solutions for the recovery and recycling of grounded tires and

discarded High Density Polyethylene (HDPE) composites.

MATERIALS AND METHODS

Materials

The polymer used in this project was sourced locally around Ahmadu Bello University Samaru Campus, Zaria, Kaduna. The polymer used were mainly HDPE waste bottles, which were washed, rinsed and dried to remove contaminants. The cleaned HDPE bottles were then crushed to reduce the size and allow melting to proceed easily.

The rubber powder obtained was separated from other particles and sieved using 150 μ , 212 μ , 300 μ and 425 μ sieve sizes at Department of Civil Engineering Ahmadu Bello University, Zaria. The sample was stored in a container prior to compounding.

Equipments

Some of the equipment's used are: Rubber powder, HDPE (waste), two roll mill machine (CPS183 North Bergen U.S.A), gigital weighing machine, Compression moulding machine (DT8679-7585 Wenzhou zingiaug chine), Changda shredding machine, Universal Material testing machine (Cat Nr. Model:261- 100KN- Enerpac), Digital Tensile testing machine (TM2101-T7), Ruler, Sieves (150 μ , 212 μ , 300 μ , 425 μ), Distilled water, DMA Machine, Scissors, Hydraulic press machine, Mold (120mm/240mm/4mm), SEM machine (JOEL JSM-7600F).

METHODS

Compounding and Compression

The composite was compounded using two-roll mill machine, which consist of two cylindrical oppositely rotating roll with an adjustable knobs, which help to increase or decrease the nip of the rollers at a rotational speed and control temperature.

The composite was prepared using the following ratios:

Table 1.1: Matrix and reinforcements composition

Samples	Amount of HDPE(g)	Size of rubber powder/ amount of rubber powder(g)
Control	150g	

150μ	135g	150μ/15g
212μ	135g	212μ/15g
300μ	135g	300μ/15g
425μ	135g	425μ/15g

The matrix (HDPE) was melted in the two rolls machine at a temperature of 180°C, then the filler was added with the aid of the rotating heated rolls and mixed together for about 20 minutes. As the materials melt the nip was reduced to make the melt form a band. Then, a knife-like object is used to remove the band from the rolls. This is done on a hydraulic press machine, which makes use of hydraulic pressure.

After blending, the band formed was fitted into a mold plate of (240mm/120mm/4mm) and wrapped with aluminium foil paper to which silicon oil was applied for easy removal of the sample after cooling. The mold was then mounted on the bottom plate with an oversized pressure valve tightened and pressure built-up using the handle, the top and bottom plates of hydraulic press was pressed against each other with the sample in-between the plates under pressure of 10MPa. The sample was pressed for about 15mins at a temperature of 220°C, then the presser was removed and placed for cooling. After cooling, the samples were removed from the aluminium foil paper and cut into different required dimensions for mechanical, thermal and morphological analysis.

CHARACTERIZATION

The following tests were carried out on the composites

Morphological Analysis

The morphology of the composite, was observed using Scanning Electron Microscopy (SEM). The ground rubber filled HDPE composite materials with varying rubber particles content were studied using a scanning electron microscope (FESEM, Joel MT, model JSM-7600F) in order to ascertain the interfacial adhesion and smooth cross section of all the specimens fabricated.

Thermal Analysis

The thermal behavior of the composite material was analysis with various specimens subjected to Dynamic Mechanical Analysis (DMA), which

applies a sinusoidal force and measures the samples response at a given temperature. The samples were studied at: Frequency/ frequencies: 2Hz, 5Hz and 10Hz, Amplitude: 60um, Heating rate: 6k/min and the End temperature: 120°C.

Mechanical Test

Tensile Strength

The tensile strength of the composite samples depends on the strength of the materials, length of the material and the material matrix interaction. This test was used to determine the tensile modulus, percentage elongation and tensile strength of the materials. The samples were cut into dumbbell shape according to the ASTM D638-98 standard. The required dimension of each sample used was 100mm×30mm×3mm. The test was carried out using Universal Materials Testing Machine (Model, TM2101-T7 10KN) at the department of Polymer and Textile Engineering, Ahmadu Bello University, Zaria. The machine provides a mechanism for applying different measurable loads to the sample with a device to measure the change in the length of the sample. The sample was attached to the machine through a clamp grip. The actual procedure is to apply the stretching load to the sample starting from a zero value to an ultimate value when the sample fractures.

$$\text{Tensile strength} = \frac{P}{A} \quad (1)$$

Where;

P = Breaking load in (N), A= Cross-sectional area of sample in (mm²)

Tensile strain is the ratio of extension of original length. It has no unit.

$$\text{Tensile strain } (\epsilon) = \frac{\text{Extended length}}{\text{Original length}} = \frac{\Delta l}{l} \quad (2)$$

Percentage elongation (elongation at break) is the ratio of extension of original length multiplied by 100.

$$\text{Elongation at break } (\%) = \frac{\Delta l}{l} \times 100 \quad (3)$$

Tensile (Young) modulus is the measure of the ability of a material to withstand changes in length

when under lengthwise tension or compression. It is also equal to the longitudinal stress divided by strain.

$$\text{Young modulus } (\gamma) = \frac{\text{Stress}}{\text{Strain}} \quad (4)$$

Flexural Analysis

The test was carried out using the Enerpac Universal Testing Machine (100KV) and measured under a three-point bending approach according to ASTM D790. The samples were cut into rectangular bar with the dimension of 60mm×30mm×4mm. The was applied along a sharp line perpendicular to the length of the bar and at a point midway the gauge length of the support. The load at the point of fracture was recorded from which

the cross-breaking strength was determined by using standard formulae. The flexural strength and modulus were calculated using the following equation;

$$\text{Flexural strength} = \frac{3FL}{2bd^2} \quad (5)$$

$$\text{Flexural modulus} = \frac{FL^3}{4bd^3D} \quad (6)$$

Where;

F = force (N), L = span length (mm), b =width (mm), d = thickness(mm), D= deflection (mm)

RESULTS AND DISCUSSION

Morphological Characterization

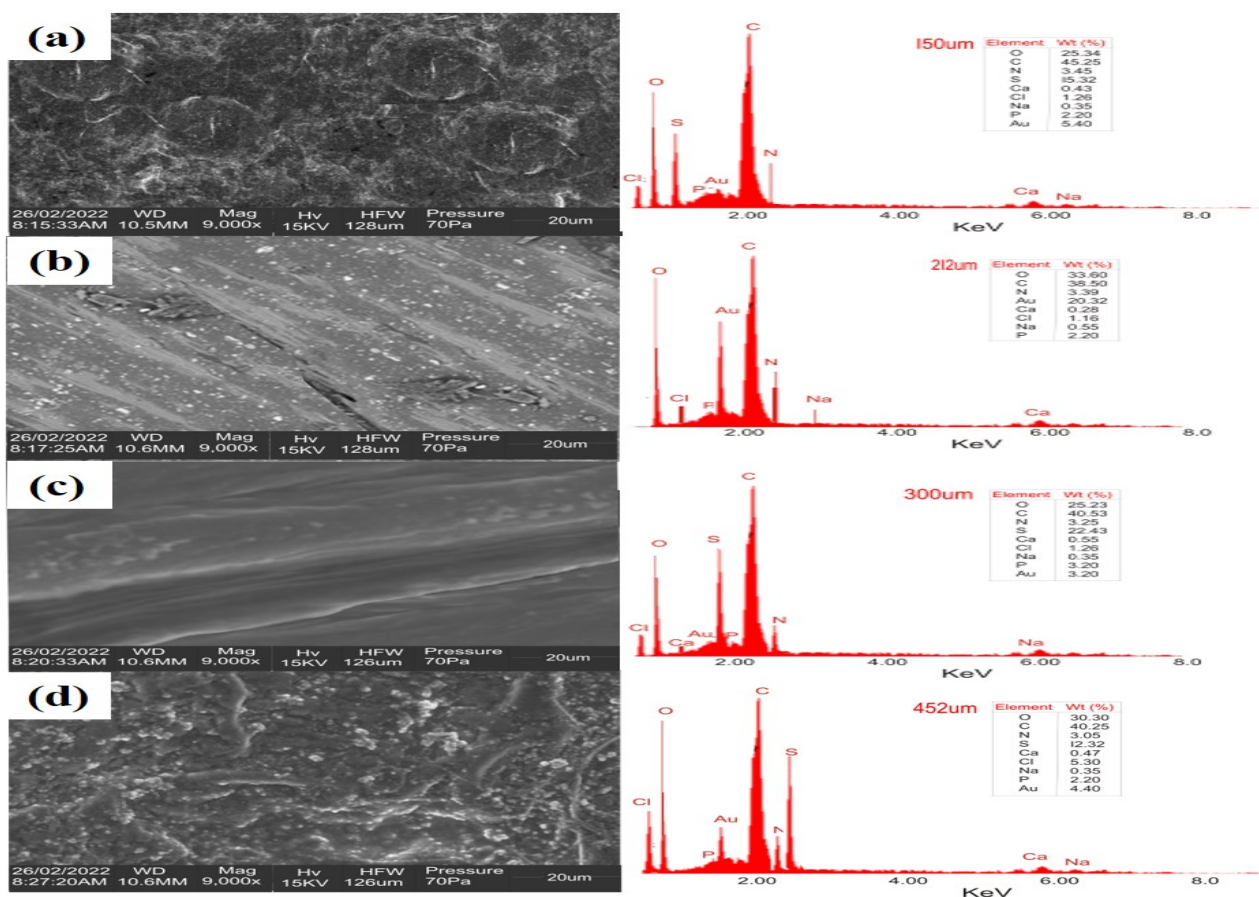


Plate 4.1: SEM image of developed HDPE filled Ground Rubber (GTR) at (a) 150μm (b) 212μm (c) 300μm (d) 425μm

SEM micrograph was used to determine the effect of different particle size of Ground Rubber on the developed HDPE composite, as illustrated in plate 4.1 above. (a) the Ground Rubber was poorly dispersed within the matrix (HDPE) which might result in improper stress transfer along the interface and formation of cracks. Hence, this might limit the interfacial adhesion within the developed composite causing reduction in mechanical strength.

Increasing the GTR concentration as shown in Plate 4.1(b) led to the agglomeration of weakly dispersed clusters and heterogenous morphology promoting voids formation at the interface attributed to incompatibility and high interfacial tension between the matrix and the filler. This led to interfacial voids, GTR pull-out from the HDPE matrix (Fazeli, Keley and Biazar, 2018). Incompatibility in the sample due the poor interfacial adhesion might cause catastrophic failure.

However, increasing the filler content from 150 μ m to 212 μ m improves the interfacial adhesion with the HDPE matrix leading to more homogenous morphology and reduction in defects like cracks, voids, agglomeration, cavity etc. Further increase in the filler concentration (from 212 μ m to 300 μ m) gave more homogenous micrograph and rough fractured surface which suggest possible effect of proper blend and uniform distribution of the filler. Proper blend suggests better interaction and proper dispersion of the GTR fillers in the matrix. Generally, an improved interfacial stress transfer is expected due to the difference in morphology. However, further addition of GTR particles up 425 μ m led to rough surface which causes

interfacial gaps between the matrix and the filler. Comparing the surfaces, no particle agglomeration was seen in Plate 4.1(c) with fewer voids. Therefore, it is expected that the developed composite with fewer void might give better mechanical properties due to more effective stress transfer and better interfacial quality (Fazli *et al.*, 2022).

Thermal Characterization

The stiffness of the polymer blends with respect to interfacial bonding and phase morphology can be understood from the storage modulus curves (Ali and Rodrigue, 2021). The maximum energy stored by the material shows how rigid the materials could be and this could be seen from the curve as illustrated in Figure 4.2. which shows the effect of different particle sizes of GRT on the HDPE/GRT blends. The control which is 150 g of pure HDPE had a higher storage modulus which is as a result of the semi-crystalline nature of HDPE structure, which also yields high stiffness. As shown in Figure 4.2 below, as the different particle sizes of the GRT filler were introduced into the rigid HDPE matrix, the storage modulus decreases except for sample made with 150 μ m, attributed to the material softening caused by the soft rubber particles which result in lower rigidity. Hence, incorporation of 300 μ m particles of GTR results in higher rigidity, ascribed to better stress transfer between the phases as a result of better interaction between matrix and the GTR. As seen in Plate 4.1, the sample with 300 μ m gave proper blend which reduce the interfacial tension between the matrix and the GTR causing higher storage modulus and better interfacial interaction

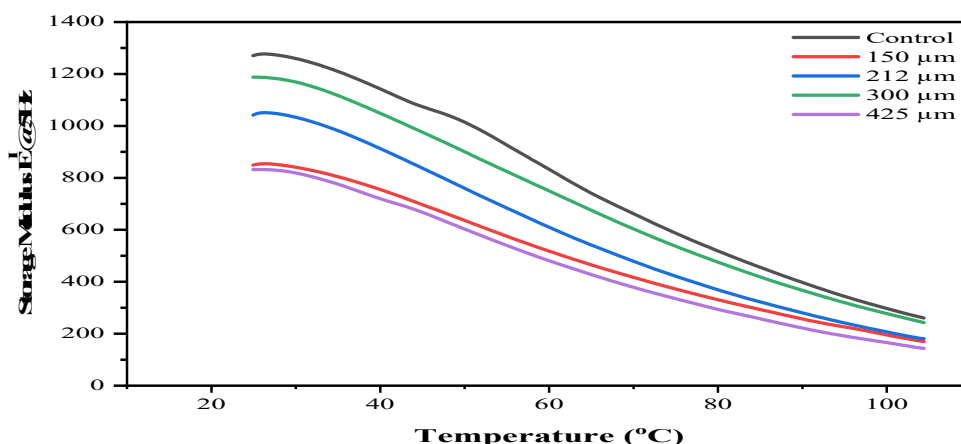


Figure 4.2: The storage modulus of the unreinforced and reinforced GTR/HDPE polymer composite as a function of temperature.

The loss tangent ($\tan \delta$) of the developed samples with variation in the GTR concentration is illustrated in Figure 4.3. Generally, it is expected that damping is affected by friction resulting from the sliding effect between the filler and matrix and also the quality of the blend interface (Zhao et al; 2019). As expected, the increase in the concentration of GTR particles led to increase in the loss modulus since the GTR particles enhances the viscoelastic property of the composite. Also,

the rise in the loss modulus is attributed the complex crosslink density and the presence of carbon black which reduce or prevent mobility within the polymer chain. Low peak intensity is observed for reinforced samples and this indicate improved compound elasticity. Hence, lower energy is required for molecular chain motion as the samples move close to rubber state from glassy state (Ali and Rodrigue, 2021).

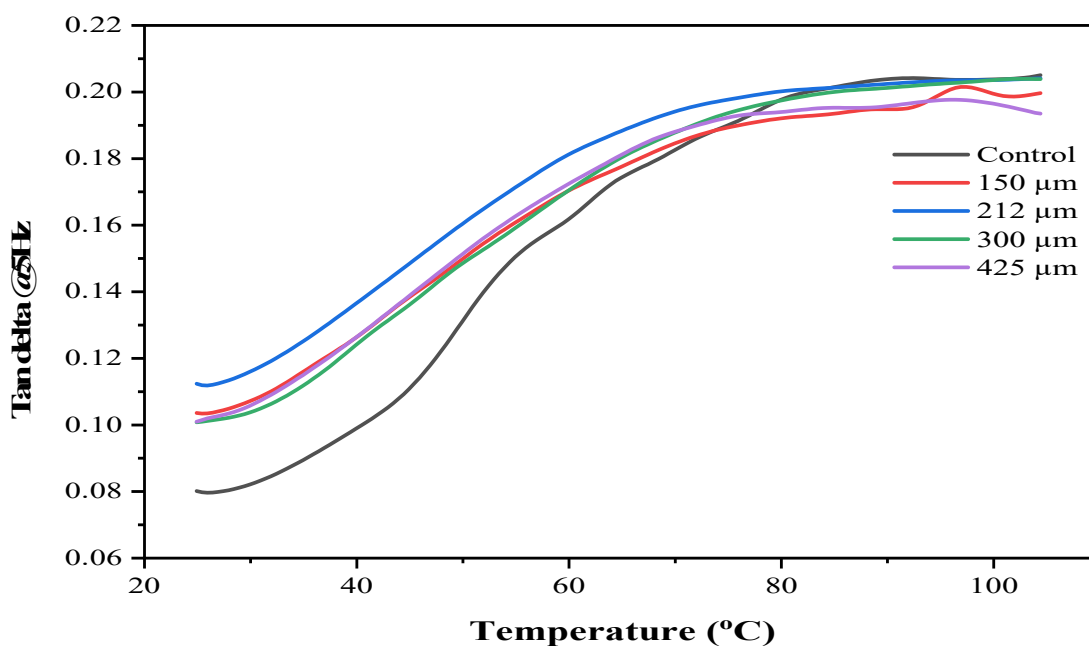


Figure 4.3: Tan delta of the unreinforced and reinforced GTR/HDPE polymer composite as a function of temperature.

Mechanical Analysis

Tensile Test

With the inclusion of GTR particles in the matrix it causes reduction in the tensile properties attributed to the constraint in the stress transfer and the weak interfacial adhesion between the filler and the matrix. Figure 4.4 shows the tensile strength and tensile modulus of the unreinforced and reinforced GTR/HDPE polymer composite. In Figure 4.4, the addition of GTR particles reduced the tensile strength of the pure HDPE matrix. For instance, incorporating 150 μm and 212 μm of GTR particles in the matrix caused the decline in the tensile strength by 28 % and 31 % respectively.

The inclusion of 212 μm particles of GTR) causes higher reduction in the tensile strength with a decline by 31 %. This could be attributed to more agglomeration as seen in Plate 4.1 and weak interfacial adhesion between the matrix and the filler causing easy initiation of crack in the sample and possible premature failure. The samples made with 300 μm and 425 μm gave the best tensile strength with 21.4 MPa and 21.5 MPa respectively. Similarly, the tensile modulus declines with the addition of the 150 μm GTR particles from 0.79 MPa to 0.71 MPa and further reduction was observed with increase in the particle size (see Figure 4.4).

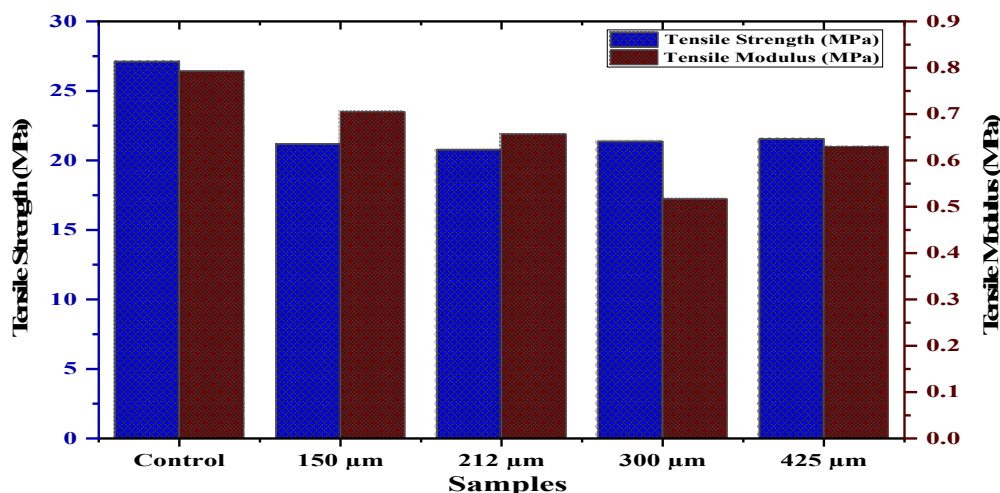


Figure 4.4 Tensile strength and Tensile modulus of the unreinforced and reinforced GTR/HDPE polymer composite

The effect of GTR particles on the elongation at break is an important parameter used to show the homogeneity and compatibility of the GTR particle/HDPE blend. Figure 4.5 illustrate the elongation at break of the produced GTR/HDPE polymer. The addition of the filler at different particle size causes decline in the elongation at break. also, the incorporation of 150 μm of the particles reduced the elongation at break from 34.1% to 30.1% the result shows that using higher particle size of GTR particles gave much higher

elongation at break but increasing the particles from 300 μm to 425 μm resulted in the decline in the elongation at break from 41.4% to 34.2%. The high elongation at break seen in samples produced with higher particle size could be attributed to high affinity, better bond and good compatibility between the blend while this reduces the stress concentration point. It is expected that the presence of the GTR particles in the matrix might serve as point of crack initiation which could have hinder the flow or mobility of the thermoplastic

molecules. Hence, the elongation at break reported in this study gave similar trend with the values reported by Wang et al., (2018). The authors reported that the elongation at break was around 50

% and it was stated that high elongation at break causes good interfacial adhesion between the blend making initiation of cracks difficult.

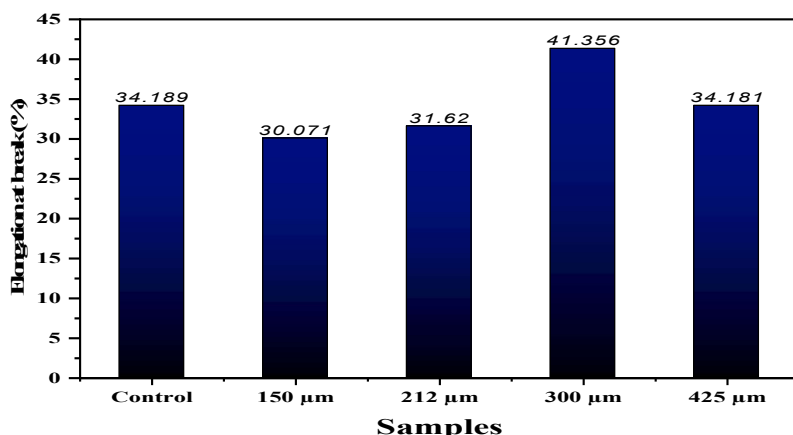


Figure4.4: Elongation at break of the unreinforced and reinforced GTR/HDPE polymer composite

Flexural Analysis

The flexural stress and modulus are shown in Figure 4.6, with continues increases of the GRT particles it causes decline in the flexural stress and modulus, having similar trend with the tensile modulus. The result obtained from this study is similar to that of Ali and Denis (2021), the addition of the GRT particles causes reduction in the flexural modulus by more than 30%. The flexural modulus dropped from 641MPa to 579MPa, when 150 μm GRT particles were first introduced resulting in 11% decline. Similar reduction can be seen for samples loaded with 212 μm, 300 μm and

425 μm of the GRT particles, which decline by 23%, 32% and 111% respectively. This decrease can be attributed to the presence of voids in the developed polymer blends. However, the developed sample with the largest particles gave the least flexural modulus attributed to the poor interfacial interaction between the filler and HDPE polymer molecules as seen in the SEM micrograph (Plate 2.1)

Hence, it could be concluded that the interfacial interaction and dispersion of the filler strongly influences the mechanical properties, and these are controlled by the particle size of the GRT filler.

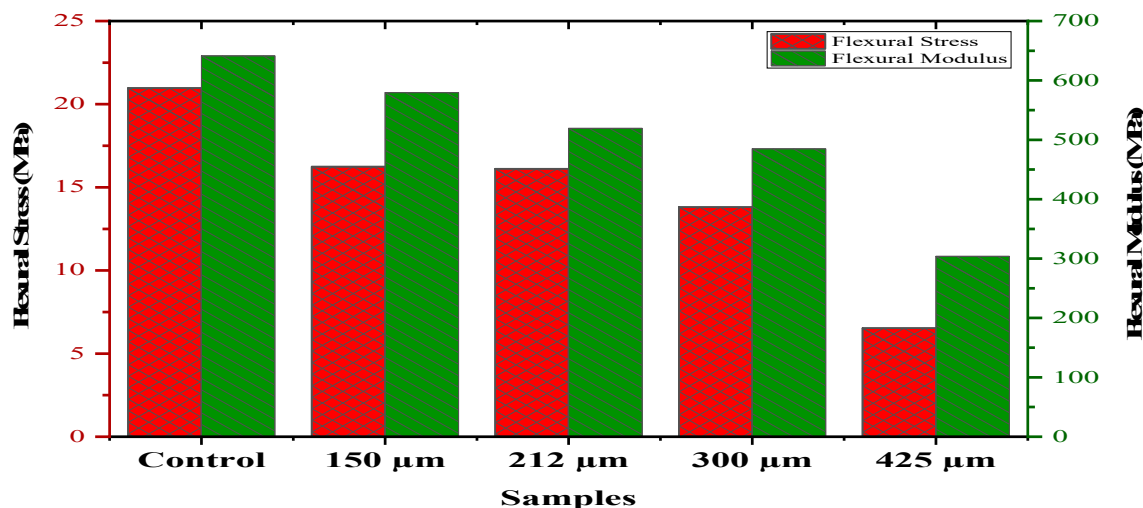


Figure 4.5: Flexural stress and Flexural modulus of the unreinforced and reinforced GRT/HDPE polymer composite

CONCLUSION

Waste GRT filler is blended with HDPE (matrix) to adhere and produce a hybrid composite with improved properties. Samples were produced by melt-mixing HDPE/GRT with varying particle sizes (150μm, 212μm, 300μm and 425μm) using a two-roll mill and a compression molding machine. Poor dispersion and defects were observed upon addition of bigger GRT particles which makes the entire material to possess poor mechanical properties as a result of their crosslinking. On the other hand, smaller particle sizes of the GRT (150 μm) with less voids gave good melting properties, higher crystallinity, as well as good tensile properties of the mixture, this is attributed to their higher specific surface area. This point can further be confirmed from the SEM Micrographs. The resulting composite with fewer voids gives better mechanical properties due to the homogeneity in particle dispersion. During the Dynamic Mechanical Analysis (DMA), it was observed that the incorporation of higher particle sized GRT (300μm) gave higher rigidity of the resulting composite. The loss tangent was deduced from the DMA result and it implied that when the size of the Ali GRT particles increases, the loss modulus increases and vice-versa. Also, the tensile strength and tensile modulus of the developed polymer composite increased according to the amount GRT particle sizes in the HDPE, 300μm and 425μm had

higher tensile strength of the among the resulting composites, this however is not applicable to the flexural modulus because, increase in the reinforcement particles led to decrease in the flexural modulus as well as the flexural stress.

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Recommendations: The authors recommended more additional test to be carried out on the Dynamic Mechanical Analysis (DMA) to study the storage modulus curves which is a function of temperature and it can help to understand the stiffness of the polymer blends with respect to phase morphology and interfacial bonding.

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